

Clinical features and surgical outcome of medial temporal lobe epilepsy with a history of complex febrile convulsions

J. Janszky^{a,b,*}, R. Schulz^b, A. Ebner^b

^a *Epilepsy Center, National Institute of Psychiatry and Neurology, Budapest, Hungary*

^b *Epilepsy Center Bethel, Bielefeld, Germany*

Received 30 January 2003; received in revised form 28 April 2003; accepted 4 May 2003

Abstract

Purpose: Temporal lobe epilepsy (TLE) is frequently associated with hippocampal sclerosis (HS) and complex febrile convulsions (CFC). The causal relationship between TLE, HS, and CFC is unclear. There is also contradictory data whether CFC-associated TLE is a distinct epilepsy syndrome and has different surgical outcome than other medial TLEs.

Methods: We investigated 133 patients (aged 16–59 years) with HS-associated TLE. Thirty-six patients with CFC (CFC group) versus 97 patients without febrile convulsions (NFC group) were compared for clinical history, video-EEG recorded seizure semiology, and surgical outcome.

Results: In the CFC group the right-sided HS (67% versus 32%) occurred more frequently than in the NFC group ($P < 0.001$). The two groups did not differ according to the clinical features, both groups share the typical symptoms and findings of the medial TLE. In the CFC group, seizure-freedom 2 years after surgery was 91%, while in the NFC group it was only 64% ($P = 0.023$). This difference was significant even after considering the other known predictive factors for medial TLE.

Conclusions: Medial TLE with CFC is not a distinct epilepsy syndrome. The surgical outcome, however, is much more favorable in these patients in comparison with medial TLE patients who had no history of febrile convulsions.

© 2003 Elsevier B.V. All rights reserved.

Keywords: Complex febrile seizures; Hippocampal sclerosis; Medial temporal lobe epilepsy; Surgical outcome; Epilepsy surgery

1. Introduction

More than 70% of patients suffering from temporal lobe epilepsy (TLE) have hippocampal sclerosis (HS) representing a distinct epilepsy syndrome: the medial TLE (Babb and Brown, 1987; Meencke and Veith,

1991; Williamson et al., 1993; French et al., 1993; Engel, 1994). More than 40% of patients with TLE have a history of febrile convulsions (Falconer, 1974; Williamson et al., 1993; Kuks et al., 1993; Janszky et al., 2003). The cause of these frequent constellations is not unequivocally clear. One of the theories explaining this relationship is that the HS is the common cause for febrile convulsions and later for the TLE (Davies et al., 1996). Kanemoto et al. (2000) hypothesized that febrile convulsions are the initial precipitating injury in the hippocampus on the basis

* Corresponding author. Present address: Epilepsie-Zentrum Bethel, Mara Krankenhaus, Maraweg 21, Bielefeld 33617, Germany. Tel.: +49-5211443697; fax: +49-5211445246.
E-mail address: janszky@index.hu (J. Janszky).

of an inherited abnormal cytokine reaction. A third theory is that a preexisting developmental abnormality may predispose to febrile convulsions that then may cause additional hippocampal damage leading to HS (Fernandez et al., 1998). The most widely accepted theory is that prolonged febrile convulsions cause the HS and the HS is responsible for the TLE appearing some years after (Maher and McLachlan, 1995; VanLandingham et al., 1998; Szabó et al., 1999; Schulz and Ebner, 2001). This theory is supported by recent experimental data: an appropriate-age rat model of febrile convulsions suggests that long duration of the febrile seizures causes transient morphological abnormalities in CA1 and CA3 subfields of the hippocampus: in the same regions which are most primary affected in the human HS (Tóth et al., 1998). Moreover, in this model prolonged febrile seizures cause a long-term seizure susceptibility in 100% (Dube et al., 2000), suggesting that the duration of complex febrile seizure (CFC) is the only cause for the hippocampal damage leading to TLE.

It is still a question whether TLE combined with a history of febrile convulsions form a separate entity. The importance of this question is not only the fact that febrile convulsions and HS are the most common epilepsy risk factors in TLE in the sense that the clinical features of the TLE with a history of febrile convulsions may have clinical and therapeutical importance; but it may also highlight the connection between TLE, febrile convulsions, and HS. Although Kanemoto et al. (1998) found that the TLE associated with CFC is a distinct entity, they did not consider whether patients had HS-associated TLE or non-lesional TLE. They found that CFC is associated with the clinical features of medial TLE. However, this may be related to the frequent association of HS with CFC. Furthermore, clinical features described in their study were not based on video-EEG monitoring. Others found that TLE together with a history of CFC associates with a favorable outcome after temporal lobe resection (So et al., 1989; Abou-Khalil et al., 1993; Kanemoto et al., 1998). These studies also ignored whether the patients had HS. Because HS is often associated with a history of CFC, it is reasonable to assume that in these patients the favorable outcome might be related to the HS and not to the CFC since TLE with HS has an excellent surgical outcome (Radhakrishnan et al., 1998).

The aim of our study is to analyze whether the medial TLE coupled with CFC has different clinical features and surgical prognosis in comparison with the medial TLE *without* a history of febrile convulsions. For this purpose, we selected only patients with HS-associated TLE who had video-EEG seizure recording.

2. Methods

2.1. Patients

In this retrospective study we included all patients who consecutively underwent presurgical evaluation at our center from 1994 to 2002 and had intractable *temporal lobe epilepsy associated with HS*. Only patients who had long-term video-EEG and registered seizures were included. Our further inclusion criteria were:

- *Complex partial seizures* (CPS) proved by ictal video-EEG.
- *Definitive interictal epileptiform discharges* (spikes or sharp waves) over one or both *temporal* regions. Two patients with extratemporal and one subject with generalized epileptiform discharges were excluded.
- *Unilateral hippocampal sclerosis* detected by high-resolution MRI. We defined HS if both hippocampal atrophy *and* increased T2 signal intensity in hippocampus were present on MRI. Patients with bilateral HS or dual pathology (HS accompanied by other epileptogenic lesion) were excluded.

2.2. Methods

The data regarding the age at the first unprovoked seizure, usual seizure frequency, presence of generalized tonic-clonic seizures, presence and type of aura, the history and type of CFC, surgery and surgical outcome were derived from medical records registered at our center. At the admission to the presurgical unit all patients and (if present) their parents were asked about the history and circumstances of febrile convulsions. CFCs were defined if it was clear from medical records or from parents' heteroanamnesis that the febrile seizures lasted more than 15 min or showed focal ictal/postictal signs, or appeared as a status epilepticus.

2.2.1. Video-EEG monitoring

Patients underwent continuous video-scalp EEG monitoring lasting >2 days. A 64-channel computerized EEG was used. The electrodes were placed according to the 10-10 system. Their placement varied individually corresponding to the suspected epileptogenic region: usually Fp1, F3, C3, P3, O1, AF7, FC5, CP5, F7, FT7, T7, TP7, P7, SP1, F9, FT9, T9, TP9 and the homologous right-sided electrodes were used. The interictal EEG samples were automatically recorded and stored on computer. In this study, the first 2 min of each computer recorded hour were reviewed and evaluated manually by visual inspections. Ictal EEG data were recorded in files separate from the interictal files. The analysis of seizure semiology and EEG data was based on the long-term video-EEG recordings. Unilateral interictal epileptiform discharges were defined if at least 95% of interictal discharges appeared over one temporal lobe.

2.2.2. MRI investigations

The examinations were made on 1.5 or 1.0 T Siemens Magnetom MR machines. Scout views with careful slices were used to avoid asymmetry, and then sagittal T1, axial T2 as well as coronal T1, T2 and proton density or FLAIR sequences perpendicular to the long axis of the hippocampus were made, which give adequate delineation of the temporal and frontal lobes.

2.2.3. Categorization of patients

There were 199 patients who met the inclusion criteria. Of 102 patients with a history of febrile convulsions, 36 patients had highly probably CFC. It was uncertain whether the remaining patients with a history of febrile convulsions had had simple or complex febrile convulsions (CFCs) mostly due to the lack of detailed medical records or heteroanamnesis. Consequently, for the further evaluation we included only those 36 patients who had had CFC and categorized them into the CFC group. Those 97 patients who had no history of febrile convulsions were categorized into the NFC group. Thus, patients with febrile convulsions but with no certain history of CFC were excluded.

2.2.4. Surgical outcome

Of 133 patients who were included in the further evaluation, 84 had temporal lobe surgery and a 2-year postoperative follow-up. Seizure-free outcome was

defined if the patient was completely free of seizures or had non-disabling auras postoperatively (Engel's category Ia and Ib). In all surgical cases, anterior temporal lobectomy was performed. In 47 patients "standard" anterior temporal lobectomy was performed consisting of resection of the temporolateral cortex in distance >3 cm from the temporal pole and the resection of the mesiotemporal structures. In the remaining 37 patients "key hole" resection was used in which the mesiotemporal resection was accompanied with a smaller temporal resection (in distance 1–2.5 cm from the temporal pole). Key-hole resection was performed in 46% of the NFC patients and in 39% of patients with CFC. This difference was not significant ($P = 0.5$).

2.2.5. Statistical methods

Two stepwise logistic regression analyses were designed. In the first, the presence of CFC was the dependent variable, while the data derived from clinical history, seizure semiology, and EEG were the independent variables in order to find out whether there was any significant difference in clinical features between the CFC and NFC groups. In the second analysis, the surgical outcome was the dependent variable, while the presence of CFC and other factors which were thought to be predictive were the independent variables. This was to establish whether the presence of CFC is predictive information regarding temporal lobe surgery.

3. Results

The mean age of patients was 33.6 (range 16–59) years. There were 71 women and 62 men. The mean age at epilepsy onset was 10.9 years. In 55 patients, the HS was on the right side, while in 78 patients it was on the left.

Table 1 shows the clinical features of the CFC and NFC groups. There was no difference with relation to the age at onset, epilepsy duration, gender, and seizure frequency. Conversely, right-sided HS (67% versus 32%, $P < 0.001$) occurred more frequently in the CFC group.

Aura types or presence of any auras showed no difference between the two groups (see Table 2). The two groups did not differ for the video-recorded seizure semiology (Table 3) or for the EEG findings (Table 4).

Table 1
General data and clinical history of patients with or without CFC

	All patients (<i>N</i> = 133)	NFC group (<i>N</i> = 97)	CFC group (<i>N</i> = 36)	Significance level
Age at onset (year) ^a	10.9 ± 8	11.1 ± 9	10.5 ± 7	NS
Duration of epilepsy (year) ^a	22.7 ± 11	23.9 ± 11	19.5 ± 10	NS
Women	71 (53%)	53 (54%)	18 (50%)	NS
Family history of epilepsy	14 (10%)	11 (11%)	3 (8%)	NS
Right hippocampal sclerosis	55 (41%)	31 (32%)	24 (67%)	<i>P</i> < 0.001

^a Mean ± S.D.

Table 2
Auras in patients with or without CFC

	All patients (<i>N</i> = 133)	NFC group (<i>N</i> = 97)	CFC group (<i>N</i> = 36)	Significance level
Lack of aura	11 (8%)	8 (8%)	3 (8%)	NS
Presence of isolated auras	74 (56%)	51 (53%)	23 (63%)	NS
Abdominal aura	82 (62%)	57 (59%)	25 (69%)	NS
Mnestic aura	16 (12%)	11 (11%)	5 (14%)	NS
Affective aura	31 (23%)	18 (19%)	13 (36%)	NS
Olfactory aura	4 (3%)	4 (4%)	0 (0%)	NS
Auditory aura	4 (3%)	3 (3%)	1 (3%)	NS
Somatosensory aura	14 (10%)	11 (11%)	3 (8%)	NS

Table 3
Seizure semiology and characteristics of patients with or without CFC

	All patients (<i>N</i> = 133)	NFC group (<i>N</i> = 97)	CFC group (<i>N</i> = 36)	Significance level
Complex partial seizures	133 (100%)	97 (100%)	36 (100%)	NS
Secondarily generalized seizures	84 (63%)	65 (69%)	19 (53%)	NS
Monthly number of seizures ^a	11.7 ± 30	12.7 ± 33	9.4 ± 11	NS
Oral automatisms	94 (71%)	66 (68%)	28 (78%)	NS
Manual automatism	105 (79%)	78 (80%)	27 (75%)	NS
Motionless stare	36 (27%)	29 (30%)	7 (19%)	NS
Dystonic posturing	54 (41%)	39 (40%)	15 (42%)	NS
Dystonia associated with contralateral hand automatism	7 (5%)	4 (4%)	3 (8%)	NS
Ictal speech	15 (11%)	8 (8%)	7 (19%)	NS

^a Mean ± S.D.

Table 4
EEG data of patients with or without CFC

	All patients (<i>N</i> = 133)	NFC group (<i>N</i> = 97)	CFC group (<i>N</i> = 36)	Significance level
Temporal interictal epileptiform discharges	133 (100%)	97 (100%)	36 (100%)	NS
Temporo-anterior interictal epileptiform discharges	112 (84%)	79 (81%)	33 (92%)	NS
Bitemporal independent interictal epileptiform discharges	39 (29%)	28 (29%)	11 (31%)	NS
Focal slowing	124 (93%)	92 (95%)	32 (89%)	NS
Generalized slowing	29 (22%)	21 (22%)	8 (22%)	NS

Of 84 patients who underwent epilepsy surgery and had >2-year follow-up, 60 patients became seizure-free (71%). Postoperatively 91% of patients with a history of CFC became seizure free, while only 64% of patients without a history of febrile seizures had a seizure free surgical outcome. A stepwise logistic regression analysis was performed to determine whether CFC, unilateral interictal epileptiform discharges, tonic-clonic seizures, perinatal insult, or side of the surgery influenced the postsurgical outcome. Only the history of CFC associated with the surgical outcome: odds ratio (OR) = 5.9, 95% confidence interval for OR = 1.26–27.7 ($P = 0.023$). Other presumably predictive factors for TLE found by recent studies did not show any influence on the postoperative outcome: (1) 74% of the 64 patients with unilateral epileptiform discharges concordant with the operated side became seizure free. (2) 69% of 55 patients with tonic-clonic seizures became postoperatively seizure free. (3) Two (67%) of three patients with a history of perinatal insult became postoperatively seizure free. Because in the CFC group the presence of right-sided HS was over-represented, we also included the side of the surgery into the analysis of postoperative outcome. The 46 patients with left-sided surgery showed no significantly different outcome in comparison with the whole patient group: 61% of them became postoperatively seizure free. Histological results were available in 80 of 84 patients. In all patients HS was confirmed. 40% of patients of CFC group and 42% of patients with NFC group had additionally global sclerosis of the mesiotemporal structures. The difference between the two groups was not significant ($P = 0.88$, Chi-square test).

4. Discussion

This is the first study that systematically analyzed the clinical features and the surgical outcome in medial TLE associated with CFC based on clinical history, MRI data, and long-term video-EEG recordings. We found that medial TLE with CFC principally showed the same clinical picture as medial TLE without CFC.

Right-sided HS occurred more frequently in the CFC group, which is in accordance with our previous study (Janszky et al., 2003). We can only speculate what causes this lateralization difference. Although

generally HS occurs more often on the left side (Janszky et al., 2003), other epileptogenic lesions, i.e. subependymal nodular heterotopia shows a strong right-sided predominance (Raymond et al., 1994), suggesting that some epileptogenic lesions and epilepsy risk factors seem to have a strong affinity to the left hemisphere, while others to the right hemisphere.

Medial TLE is characterized by CPS beginning in childhood, frequent presence of auras which are predominantly abdominal, and anterotemporal interictal spikes (French et al., 1993; Williamson et al., 1993; Ebner, 1994; O'Brien et al., 1996). In this study, we systematically evaluated the CFC and NFC groups as to whether there were any differences in favor or in contrast to the typical medial TLE syndrome.

All patients had CPS. The age at onset showed no difference in the two groups. Auras occurred in equal frequency in both groups. Auditory and somatosensory aura, which is relatively frequent in neocortical (lateral) TLE (Ebner, 1994; Wunderlich et al., 2000), was equally infrequent in both groups. Both ictal dystonia (O'Brien et al., 1996) and ictal dystonia accompanied by contralateral unilateral hand automatisms (Dupont et al., 1999) are characteristics of medial TLE rather than lateral TLE. These ictal phenomena also occurred equally in both groups. Some authors found that the motionless stare during CPS might be more characteristic to neocortical TLE (Pacia et al., 1996). Regarding our patients, it occurred in CFC and NFC groups in equal frequency. Antero-temporal spikes (Williamson et al., 1993) occurring more often in mesial than in lateral TLE appeared equally frequently in both groups of our study. Bilateral independent epileptiform discharges which were found to be infrequent in lateral TLE in comparison with medial TLE (Pacia et al., 1996) also occurred in the same frequency in CFC and NFC groups in our study.

Some studies found that patients with CPS and a history of CFC are usually resistant to drug therapy (Schmidt et al., 1985), but have an excellent outcome after temporal lobe resection (So et al., 1989; Abou-Khalil et al., 1993; Kanemoto et al., 1998). The resistance to drug therapy (Semah et al., 1998) and more favorable surgical outcome is also true for TLE patients with HS (Berkovic et al., 1995; Radhakrishnan et al., 1998).

It is reasonable to assume that in patients with CFC, the favorable surgical outcome might be related to the

HS and not to the CFC. Indeed, more recent studies did not find a predictive value of febrile convulsions (Radhakrishnan et al., 1998; Kilpatrick et al., 1999; Hennessy et al., 2000). We were not able to test the response to antiepileptic drugs in CFC versus NFC groups because all patients suffered from intractable TLE. The outcome after temporal lobe surgery was significantly favorable in patients with a history of CFC than in patients who had no febrile seizures. Postoperatively 91% of patients with a history of CFC became seizure-free, while only 64% of patients without a history of febrile seizures had a seizure-free surgical outcome. In assessing the prognostic value of surgical outcome, we controlled all reported variables which may predict the surgical outcome by routine presurgical evaluation of medial TLE patients with HS (Rosenow and Lüders, 2001; Ebner, 2001). The two most consistently reported predictive factors for TLE surgery are the presence of a circumscribed abnormality on the MRI and unilateral spike focus (Berkovic et al., 1995; Holmes et al., 1997; Radhakrishnan et al., 1998; Schulz et al., 2000). Although in our previous prospective study we found that bilateral seizure activity (contralateral seizure spread or independent bilateral seizure activity) also associated with an unfavorable surgical outcome in TLE (Schulz et al., 2000), in this study, we did not control this variable because seizure propagation was not documented in many patients. Hennessy et al. (2000) investigating the surgical outcome of medial TLE with HS found that the absence of generalized tonic–clonic seizures, interictal EEG localization to the operated temporal lobe and the absence of perinatal brain insult correlated with favorable surgical outcome. The interictal EEG localization other than the operated temporal lobe was not investigated in our study in general because we excluded all patients with extratemporal or generalized spikes, thus we could investigate only the bitemporal versus unitemporal interictal EEG localization in CFC versus NFC groups. Since all of our patients had HS, in the multivariate analysis, we included only the presence of unilateral spike focus, secondarily generalized seizures, the history of perinatal insult, and the side of the lobectomy. None of these variables correlated with the surgical outcome. Although generalized seizures and the history of perinatal insult as predictive factors were found in a study conducted by Hennessy et al. (2000) which was not based on modern presurgical

investigation protocol (many patients had no MRI and no video-EEG monitoring in that study), the fact that unilateral spike focus had no predictive value was surprising, and may be attributed to a selection bias. After the unilateral irritative zone was reported as a strong positive prognostic factor (Holmes et al., 1997; Radhakrishnan et al., 1998), while, we predicted reduced likelihood for seizure freedom in counseling our TLE patients with bitemporal spikes. Consequently, some patients might have refused the operation for this reason. Conversely, the selection criteria for our study (i.e. excluding patients with febrile convulsions in whom we could not decide whether they had had a CFC or a simple febrile convulsion, not including patients with extratemporal/generalized spikes), may also influence the prognostic value of all other variables other than CFC (i.e. unilateral spikes, generalized seizures, and perinatal insult). Moreover, the number of patients with perinatal insult (three patients) was extremely small in our population, thus this variable had “no chance” of reaching statistical significance. However, our primary goal was not to test the predictive value of the above-mentioned variables but to focus on testing whether the predictive value of CFC is independent of these variables, and this has been proved in our study.

Three recent studies did not find febrile seizures as a predictive factor in TLE surgery (Radhakrishnan et al., 1998; Kilpatrick et al., 1999; Hennessy et al., 2000). This may be due to the fact that Radhakrishnan et al. (1998) and Kilpatrick et al. (1999) did not distinguish between CFC and other febrile seizures. Although Hennessy et al. (2000) handled CFC as a separate variable, they also included patients with febrile convulsions without known complex features in their multivariate analysis. Consequently, some patients with a history of CFC might have been categorized in the non-CFC group in their study, as the nature of febrile seizures cannot always be abstracted from medical records or during the patient history in most cases. In adult patients—at least according to our experience—it is difficult to obtain enough clinical data on the nature of febrile convulsions, and this was the reason why we excluded those patients with febrile convulsions in whom the complex nature was uncertain. Conversely, in patients with a history of CFC who had focal signs, Hennessy et al. (2000) found 92% had postoperative seizure freedom compared to the whole patient group

where 67% of patients became postoperatively seizure free, which results are in accordance with ours. Due to the small number of patients, however, this difference was not found to be significant in their study.

In conclusion, our data suggest that medial TLE with CFC principally shows the same clinical picture as medial TLE without CFC, and is probably not a well-defined, separate syndrome. We suggest that CFC may exert no direct effect on the symptoms and findings of medial TLE, only on the development of HS. Conversely, a history of complex febrile seizures predicts a favorable surgical outcome, which may be the consequence of a more circumscribed epileptogenic region, a more specific hippocampal damage (Tóth et al., 1998) compared to patients in whom the HS evolved after other initial precipitating injuries.

Acknowledgements

This work was supported by a grant from the *Deutsche Forschungsgemeinschaft (DFG-Eb 111/2-2)* and from the *Humboldt Stiftung* (Dr. Janszky).

References

- Abou-Khalil, B., Andermann, E., Andermann, F., Olivier, A., Quesney, L.F., 1993. Temporal lobe epilepsy after prolonged febrile convulsions: excellent outcome after surgical treatment. *Epilepsia* 34, 878–883.
- Babb, T.L., Brown, W.J., 1987. Pathological findings in epilepsy. In: Engel, J. (Ed.), *Surgical Treatment of Epilepsies*. Raven Press, New York, pp. 511–540.
- Berkovic, S.F., McIntosh, A.M., Kalnins, R.M., 1995. Preoperative MRI predicts outcome of temporal lobectomy. *Neurology* 45, 1358–1363.
- Davies, G.K., Hermann, B.P., Dohan, F.C., Foley, K.T., Bush, A.J., Wyler, A.R., 1996. Relationship of hippocampal sclerosis to duration and age of onset of epilepsy, and childhood febrile seizures in temporal lobectomy patients. *Epilepsy Res.* 24, 119–126.
- Dube, C., Chen, K., Eghbal-Ahmadi, M., Brunson, K., Soltesz, I., Baram, T.Z., 2000. Prolonged febrile seizures in the immature rat model enhance hippocampal excitability long term. *Ann. Neurol.* 47, 336–344.
- Dupont, S., Semah, F., Boon, P., 1999. Association of ipsilateral motor automatisms and contralateral dystonic posturing: a clinical feature differentiating medial from neocortical epilepsy. *Arch. Neurol.* 56, 927–932.
- Ebner, A., 1994. Lateral (neocortical) temporal lobe epilepsy. In: Wolf, P. (Ed.), *Epileptic Seizures and Syndromes*. John Libbey Ltd., London, pp. 375–382.
- Ebner, A., 2001. Preoperative evaluation in epilepsy surgery: some principal considerations. In: Lüders, H., Comair, Y.G. (Eds.), *Epilepsy Surgery*. Lippincot-Raven Publishers, Philadelphia, pp. 177–183.
- Engel, J., 1994. Bilateral temporal lobe epilepsy. In: Wolf, P. (Ed.), *Epileptic Seizures and Syndromes*. John Libbey Ltd., London, pp. 359–368.
- Falconer, M.A., 1974. Mesial temporal (Ammon's horn) sclerosis as a common cause of epilepsy: aetiology, treatment, and prevention. *Lancet* 2 (7883), 767–770.
- Fernandez, G., Effenberger, O., Vinz, B., Steinlein, O., Elger, C.E., Dohring, W., Heinze, H.J., 1998. Hippocampal malformation as a cause of familial febrile convulsions and subsequent hippocampal sclerosis. *Neurology* 50, 909–917.
- French, J.A., Williamson, P.D., Thadani, V.M., Darcey, T.M., Mattson, R.H., Spencer, S.S., Spencer, D.D., 1993. Characteristic of medial temporal lobe epilepsy. I. Results of history and physical examination. *Ann. Neurol.* 34, 774–780.
- Hennessy, M.J., Elwes, R.D.C., Rabe-Hesketh, S., Binnie, C.D., Polkey, C.D., 2000. Prognostic factors in the presurgical treatment of medically intractable epilepsy associated with mesial temporal sclerosis. *Acta Neurol. Scand.* 103, 344–355.
- Holmes, M.D., Dodrill, C.B., Ojemann, G.A., Wilensky, A.J., Ojemann, L.M., 1997. Outcome following surgery in patients with bitemporal interictal discharges. *Neurology* 48, 1037–1040.
- Janszky, J., Woermann, F.G., Barsi, P., Schulz, R., Halász, P., Ebner, A., 2003. Right-sided hippocampal sclerosis is more common than left after febrile convulsions. *Neurology* 60, 1209–1210.
- Kanemoto, K., Takuji, N., Kawasaki, J., Kawai, I., 1998. Characteristic and treatment of temporal lobe epilepsy with a history of complicated febrile convulsions. *J. Neurol. Neurosurg. Psychiatry* 64, 245–248.
- Kanemoto, K., Kawasaki, J., Miyamoto, T., Obayashi, H., Nishimura, M., 2000. Interleukin (IL)1 β , IL-1 α , and IL-1 receptor antagonist gene polymorphisms in patients with temporal lobe epilepsy. *Ann. Neurol.* 47, 571–574.
- Kilpatrick, C., Cook, M., Matkovic, Z., O'Brien, T., Kaye, A., Murphy, M., 1999. Seizure frequency and duration of epilepsy are not risk factors for postoperative seizure outcome in patients with hippocampal sclerosis. *Epilepsia* 40, 899–903.
- Kuks, J.B.M., Cook, M.J., Fish, D.R., Stevens, J.M., Shorvon, S.D., 1993. Hippocampal sclerosis in epilepsy and childhood febrile seizures. *Lancet* 342, 1391–1394.
- Maher, J., McLachlan, R.S., 1995. Febrile convulsions—is seizure duration the most important predictor of temporal lobe epilepsy? *Brain* 118, 1521–1528.
- Meencke, H.J., Veith, G., 1991. Hippocampal sclerosis in epilepsy. In: Lüders, H. (Ed.), *Epilepsy Surgery*. Raven Press, New York, pp. 705–715.
- O'Brien, T.J., Kilpatrick, C., Murrie, V., Vorgan, S., Morris, K., Cook, M.J., 1996. Temporal lobe epilepsy caused by medial temporal sclerosis and temporal neocortical lesions. A clinical and electroencephalographic study of 46 pathological lesions. *Brain* 119, 2133–2141.
- Pacia, S.V., Devinsky, O., Perrine, K., Ravdin, L., Luciano, D., Vazquez, B., Doyle, W.K., 1996. Clinical features of neocortical temporal lobe epilepsy. *Ann. Neurol.* 40, 724–730.

- Radhakrishnan, K., So, E.L., Silbert, P.L., Jack, C.R., Cascino, G.D., Sharbrough, F.W., O'Brien, P.C., 1998. Predictors of outcome of anterior temporal lobectomy for intractable epilepsy—a multivariate study. *Neurology* 51, 465–471.
- Raymond, A.A., Fish, D.R., Stevens, J.M., Sisodiya, S.M., Alsanjari, N., Shorvon, S.D., 1994. Subependymal heterotopia: a distinct neuronal migration disorder associated with epilepsy. *J. Neurol. Neurosurg. Psychiatry* 57, 1195–1202.
- Rosenow, F., Lüders, H.O., 2001. Presurgical evaluation of epilepsy. *Brain* 124, 1683–1700.
- Schmidt, D., Tsai, J.J., Janz, D., 1985. Febrile seizures in patients with complex partial seizures. *Acta Neurol. Scand.* 72, 68–71.
- Schulz, R., Ebner, A., 2001. Prolonged convulsions and mesial temporal lobe epilepsy in an identical twin. *Neurology* 57, 318–320.
- Schulz, R., Lüder, H.O., Hoppe, M., Tuxhorn, I., May, T., Ebner, A., 2000. Interictal EEG and ictal scalp EEG propagation are highly predictive of surgical outcome in mesial temporal lobe epilepsy. *Epilepsia* 41, 564–570.
- Semah, F., Picot, M.C., Adam, C., Broglin, D., Arzimanoglou, A., Bazin, B., Cavalcanti, D., Baulac, M., 1998. Is the underlying cause of epilepsy a major prognostic factor for recurrence? *Neurology* 51, 1256–1262.
- So, N., Olivier, A., Andreemann, F., Gloor, P., Quesney, L.F., 1989. Results of surgical treatment in patients with bitemporal epileptiform abnormalities. *Ann. Neurol.* 25, 432–439.
- Szabó, C.Á., Wyllie, E., Siavalas, E.L., Najm, I., Ruggieri, P., Kotagal, P., Lüders, H., 1999. Hippocampal volumetry in children with or without complex febrile seizures. *Epilepsy Res.* 33, 1–9.
- Tóth, Z., Yan, X., Haftoglou, S., Ribak, C.E., Baram, T.Z., 1998. Seizure-induced neuronal injury: vulnerability to febrile seizures in an immature rat model. *J. Neurosci.* 18, 4285–4294.
- VanLandingham, K.E., Heinz, E.R., Cavazos, J.E., Lewis, D.V., 1998. MRI evidence of hippocampal injury after prolonged, focal febrile convulsions. *Ann. Neurol.* 43, 413–426.
- Willamson, P.D., French, J.A., Thadani, V.M., Kim, J.H., Novelly, R.A., Spencer, S.S., Spencer, D.D., Mattson, R.H., 1993. Characteristics of medial temporal lobe epilepsy. II. Interictal and ictal scalp electrography, neuropsychological testing and neuroimaging. *Ann. Neurol.* 34, 781–787.
- Wunderlich, G., Schüller, M.F., Ebner, A., Holthausen, H., Tuxhorn, I., Witte, O.W., Seitz, J.R., 2000. Temporal lobe epilepsy with sensory aura: interictal glucose hypometabolism. *Epilepsy Res.* 38, 139–149.